

## Preface

**Ernst Ahlberg**

*Mölnlycke Healthcare, Sweden*

ERNST.AHLBERG@MOLNLYCKE.COM

**Ulf Johansson**

*Jönköping University, Sweden*

ULF.JOHANSSON@JU.SE

**Henrik Boström**

*KTH Royal Institute of Technology*

BOSTROMH@KTH.SE

**Alberto Carlevaro**

*CNR-IEIIT, Italy*

ALBERTO.CARLEVARO@AITEK.IT

**Johan Hallberg Szabadváry**

*Jönköping University and Stockholm University, Sweden*

JOHAN.HALLBERG.SZABADVARY@JU.SE

**Lars Carlsson**

*Jönköping University, Sweden*

LARS.CARLSSON@JU.SE

This volume presents the proceedings of the Fifteenth Symposium on Conformal and Probabilistic Prediction with Applications (COPA 2026). The symposium takes place at Mölnlycke Healthcare, Göteborg on September 2–4, 2026.

Overall, 47 full papers and 20 extended abstracts have been accepted for publication in the Proceedings of Machine Learning Research, Volume 329. The full papers are divided into the following eight topics:

- **Foundations and Theory** (9 papers)
  - *Aggregation in conformal e-classification* by Vladimir Vovk
  - *Inductive Venn–Abers and related regressors* by Ivan Petej and Vladimir Vovk
  - *Distortion and Consistency in Conformal Prediction* by Ilia Nouretdinov
  - *Skew-adaptive conformal prediction* by Paulo C. Marques F. and Helton Graziadei
  - *Target-Aligned Full Conformal Bayes under Continuous Label Shift* by Juyeon Kim, Hyeonsu Lee, Erkhembayar Jadamba, Seungjin Choi and Hyunjin Shin
  - *Conformal Prediction via Transported Beta Laws* by Thiago R. Ramos, Helton Graziadei and Luben M. C. Cabezas
  - *Online monotone density estimation and log-optimal calibration* by Rohan Hore, Ruodu Wang and Aaditya Ramdas
  - *Conformal prediction or the good old DKW from the 50s?* by Chancellor Johnstone and Eugene Ndiaye
  - *Corrected Max-Rank for Finite-Sample Conformal Prediction* by Filip Schlembach, Evgueni Smirnov and Mark H. M. Winands
- **Calibration, Regression and Predictive Distributions** (7 papers)

- *Well-calibrated probabilities for prediction sets* by Oskar Gustafsson, John Pavia, Styrbjörn Käll, Johan Hallberg Szabadváry, Jens Lundström and Ernst Ahlberg
- *CRPS-Optimal Binning for Univariate Conformal Regression* by Paolo Toccaceli
- *Mondrian Conformal Regressors with Overlapping Categories* by Henrik Boström and Ulf Johansson
- *Weighted Conformalized Quantile Regression for Active Learning* by Niclas Ståhl, Cecilia Sönströd and Ulf Johansson
- *Scaling-Score Conformal Prediction for Multi-Target Regression* by Sylvain Rousseau and Soundouss Messoudi
- *Conformal Calibration for Multi-Modal Regression with Missing Modalities* by Ilia Azizi
- *When Post-Hoc Calibration Hurts: Platt Scaling and Isotonic Regression on Modern Tabular Models* by Valery Manokhin and Daniel Grønhaug
- **Explainability and Decision Making** (11 papers)
  - *Turning Feature Attributions into Sufficient Explanations Using Conformal Prediction* by Amr Alkhatib and Stephanie Lowry
  - *Beyond the Predicted Class: Calibrated Explanations for Real Multiclass Decisions\** by Ayham Hanna and Tuwe Löfström
  - *Guarded Explanations: Conformal-Style Filtering for Distribution-Aware Rule Conditions* by Tuwe Löfström, Anders Hjort and Helena Löfström
  - *Performance Estimation in Hybrid Interpretable Models with Venn Predictors* by Alberto García-Galindo, Marcos López-De-Castro, Ruben Armañanzas, Niclas Ståhl and Tuwe Löfström
  - *Risk-Averse Evaluation of Conformal Scores through Conditional Value-at-Risk* by Sara Narteni, Alberto Carlevaro, Fabrizio Dabbene and Maurizio Mongelli
  - *Venn Predictive Decision-Making* by Johan Hallberg Szabadváry and Lars Carlsson
  - *TIE-BCPNN: Selective Conformal Risk Control with Audit-Trail and Counterfactual Diagnostics on Atomic-Posterior Bayesian Classifiers* by Diego Argüello Ron, Christyan Cruz Ulloa, Naresh Ravichandran, Anders Lansner, Pawel Herman, Jaime del Cerro and Oscar García Perales
  - *Explaining Conformal Prediction: Diagnosing Reliability through Feature-Conditioned Outcomes and p-Value Margins* by Antonio Caparrini, Miller-Janny Ariza-Garzón and Javier Arroyo
  - *Conformal Selection of Counterfactual Explanations: From Minimal to Robust Recourse* by Ulf Johansson, Aicha Maalej, Niclas Ståhl and Henrik Boström
  - *When Is Conformal Coverage Free? Switching Thresholds for Predict-then-Optimize* by Chandra Dronavajjala
  - *Conformal Repair on a Budget: An Empirical Protocol for Retraining Decisions* by Helena Löfström

- **Online Testing, Change and Anomaly Detection** (5 papers)
  - *Detecting Label Shift for Large Language Models with Conformal Test Martingales* by Henrik Boström
  - *Conformal Anomaly Detection in Python: Moving Beyond Heuristic Thresholds with nonconform* by Oliver Hennhöfer, Maximilian Kirsch and Christine Preisach
  - *Multivariate Change-Point Detection Using Feature-Based Inductive Conformal Martingales* by Charalambos Eliades and Harris Papadopoulos
  - *Detecting Adversarial Attacks in Image Sequences with Conformal Test Martingales* by Genghua Dong, Ziyun Li and Henrik Boström
  - *Conformalized Time Series Anomaly Thresholding with Latent Space Features* by Nancy Xu and Henrik Boström
- **Multi-Output and Structured Prediction** (4 papers)
  - *MuLaConf: Multi-label Conformal Prediction* by Kostas Katsios and Harris Papadopoulos
  - *Generating Uncertainty Sets Using Conformal Prediction for Robust Optimization in Minimum Cost Flow* by Zion Johnson, Bruce Cox and Chancellor Johnstone
  - *Lightweight Online Multivariate Conformal Prediction* by Abd Al-Rahman Hourani, Soundouss Messoudi, Sylvain Rousseau and Stefano Masi
  - *Efficient Multi-Label Conformal Classification* by Henrik Boström and Ulf Norinder
- **Robustness, Fairness and Distribution Shift** (3 papers)
  - *Fair Conformal Prediction for Individuals and Subgroups in Natural and Adversarial Settings: Theoretical Guarantees and Impossibility Results* by Alberto Carlevaro, Luca Oneto, Massimo Massa, Davide Anguita and Fabio Roli
  - *A Deployment Audit of Release-Side Risk in Conformal Triage under Prevalence Shift* by Chengze Li, Xiao Liu, Hanrong Zhang, Haiyang Peng, Yanghao Ruan, Huanhuan Ma, Chunyu Miao, Qichao Zhou, Xiangrong Qi and Philip Yu
  - *Particle-Based Conformal Prediction for Contact-Aware Uncertainty Calibration in Stratified Configuration Spaces* by Luís Marques, Kristian Popov and Dmitry Berenson
- **Applications** (8 papers)
  - *Continuous Assurance of Digital Assets via Conformal Prediction over Polytope-Defined Operating Regions* by Hee Jong Park, Kristoffer Skare, Cesar Augusto Ramos de Carvalho and Claas Rostock
  - *When to Trust Simulated Data: Conformal Prediction for Selective Synthetic Data Integration* by Sakir Furkan Yondem, Seyda Yoncaci, Gizem Karagoz, Benedikt Schlereth-Groh, Engin Avci, Ramin Tavakoli Kolagari and Fatima Rabbia Yapicioglu

- *Probabilistic Object Detection with Conformal Prediction* by Christopher Ries, Moussa Kassem Sbeyti, Nicolas Bianco and Nadja Klein
- *Improving the Sensitivity of Gravitational Wave Detection with Weighted Conformal Prediction* by Ann-Kristin Malz, Gregory Ashton and Nicolo Colombo
- *Trustworthy AI in Wound Care Documentation* by Oskar Gustafsson, Jens Lundström, Ulf Johansson, John Pavia, Niclas Ståhl, Cecilia Sönströd and Ernst Ahlberg
- *TRACE-CRC: Trajectory-Adaptive Conformal Risk Control for Multi-Step Channel State Information Prediction* by Kiarash Rezaei, Mehdi Sattari, Javad Aliakbari, Tommy Svensson, Paolo Monti and Carlos Natalino
- *Pixelwise Split Conformal Prediction for Global Temperature Emulation via Leave-One-Out Ensemble Distillation* by Bünyamin Korkut, Yunus Emre Karaoğlu, Ozan Çetin, Yusuf H. Sahin and Alper Ünal
- *Uncertainty-aware Decision Support in Power Grid Demand Forecasting* by Tuwe Löfström, Johan Hallberg Szabadváry, Kristoffer Pettersson, Madalina Aldea and Helena Löfström

• **Extended Abstracts** (20 papers)

- *Zero-Shot Probabilistic Stock Returns Forecasting with Pretrained RVFL Networks* by T. Moudiki
- *Adaptive Conformal Prediction for Full-Scale Plant pH Forecasting under Process Drift* by Jongyeop Lee and Jong Min Lee
- *Towards Group-Conditional Coverage Guarantees via Group-Invariant Representations* by Paul Melki, Lionel Bombrun and Jean-Pierre Da Costa
- *A Machine-Checked Anytime-Valid Confidence Sequence by the Method of Mixtures* by Robert Sneiderman
- *Adaptive Conformal Prediction for 5G Link Adaptation under Distribution Shift* by Emanuele Minotti, Genghua Dong and George Koudouridis
- *Finite-Sample-Corrected Conformal Test Martingales for Wireless Link Adaptation* by Peixuan Wu, George Koudouridis and Genghua Dong
- *Martingale Detection of Treatment Effects in Wound Care* by Oskar Gustafsson, John Pavia, Daniel Tsang and Ernst Ahlberg
- *Reliable Environmental Planning at HS2 Construction Sites* by Jasmin Ahluwalia, Xu Feng, Khuong An Nguyen and Matthew Tye
- *What Can Conformal Risk Control Certify for PET/CT Tumour Segmentation?* by Weiyue Zheng and Surajit Ray
- *Geometry-Aware Conformal Decoding for Motor BCI Systems* by Roberto Bonini, Zhiyuan Luo, Matteo Filippini and Patrizia Fattori
- *A Gibbs-Boltzmann Approach to Conformal Prediction under Distribution Shift* by Shivaprasad Channapura Sudhakara and Ilia Nouredinov

- *Too Good to Conform: The Inverse Quality Paradox in Counterfactual Filtering* by Fatima Rabia Yapicioglu, Abhishek Srinivasan, Juan Carlos Andresen and Henrik Boström
- *Conformal Score Prediction in Figure Skating Competitions* by Varvara Kisel, Ilia Nouretdinov and Alex Gammerman
- *No Free Lunch in Few-Step Flow Matching for Conformal Prediction* by Ziyun Li and Henrik Boström
- *Attune, Don't Prune: A Conformal Framework for Fact Preservation in News Content Attunement* by Alice Evelyn Ashby, Mai Nguyen, Zhiyuan Luo and Khuong An Nguyen
- *Label-Free Distribution Shift Detection in Truck Mode Classification via Uncertainty-Gap Monitors* by Bek-Myrza Nurmatov, Valery Manokhin and Jens Heger
- *Phase-Aware Conformal Risk Control for Selective Antimicrobial Susceptibility Display* by Yaser Kord, Moritz Hess and Jens Timmer
- *ConfBatch: Conformal Prediction for Adaptive Batch Sizing* by Anna Chatzipapadopoulou, Stavros Toumpis and John Pavlopoulos
- *Conformal Poset Prediction for Label Ranking* by Yusuf Sale and Eyke Hüllermeier
- *Robust Model Predictive Control via Conformal Prediction* by Felix Czaja, Daniil Kazantsev and Eyke Hüllermeier

The symposium features two keynote presentations:

- *Accelerating Drug Design with ML/AI* by Ola Engkvist, AstraZeneca
- *Prediction models for improving cancer diagnostics and treatment* by Martin Eklund, Karolinska Institutet

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